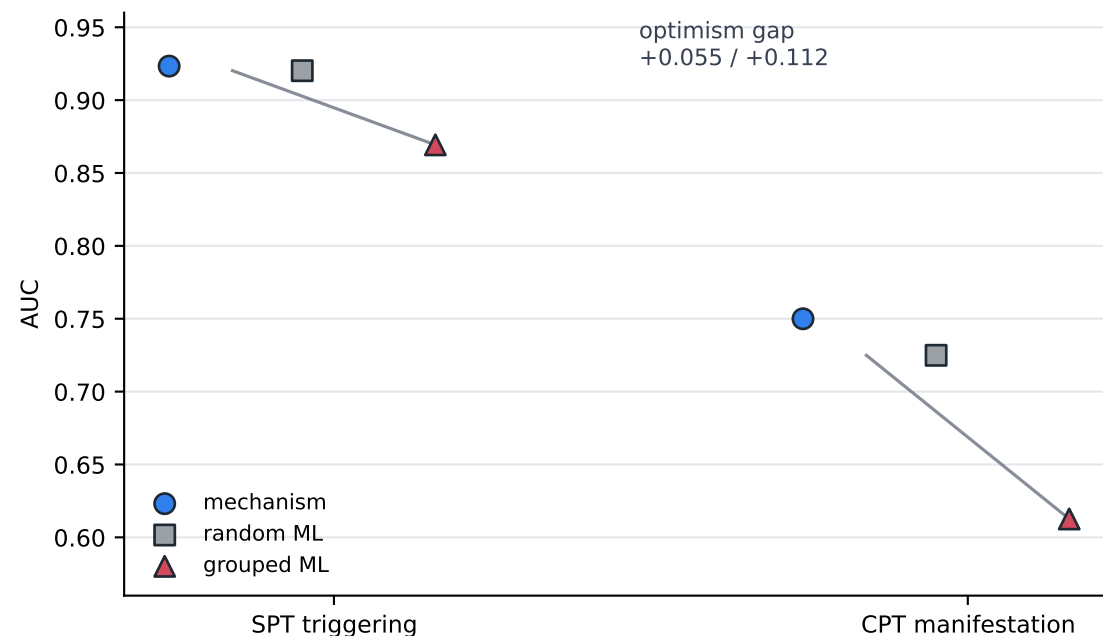
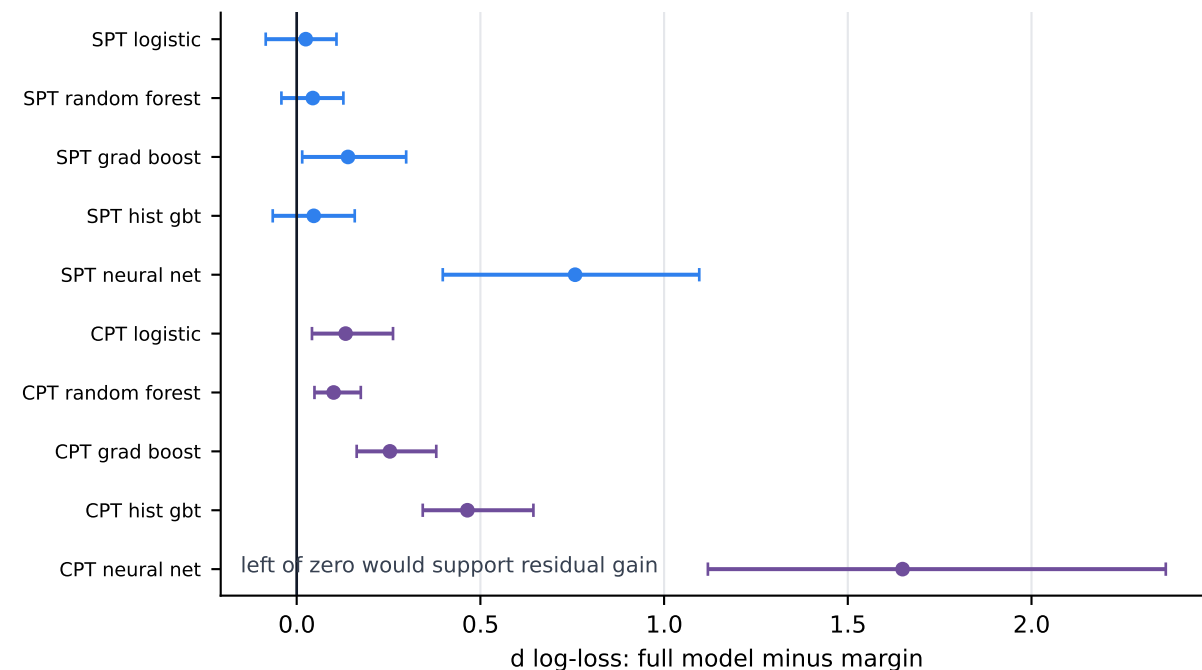


Transferable sufficiency under earthquake-grouped validation

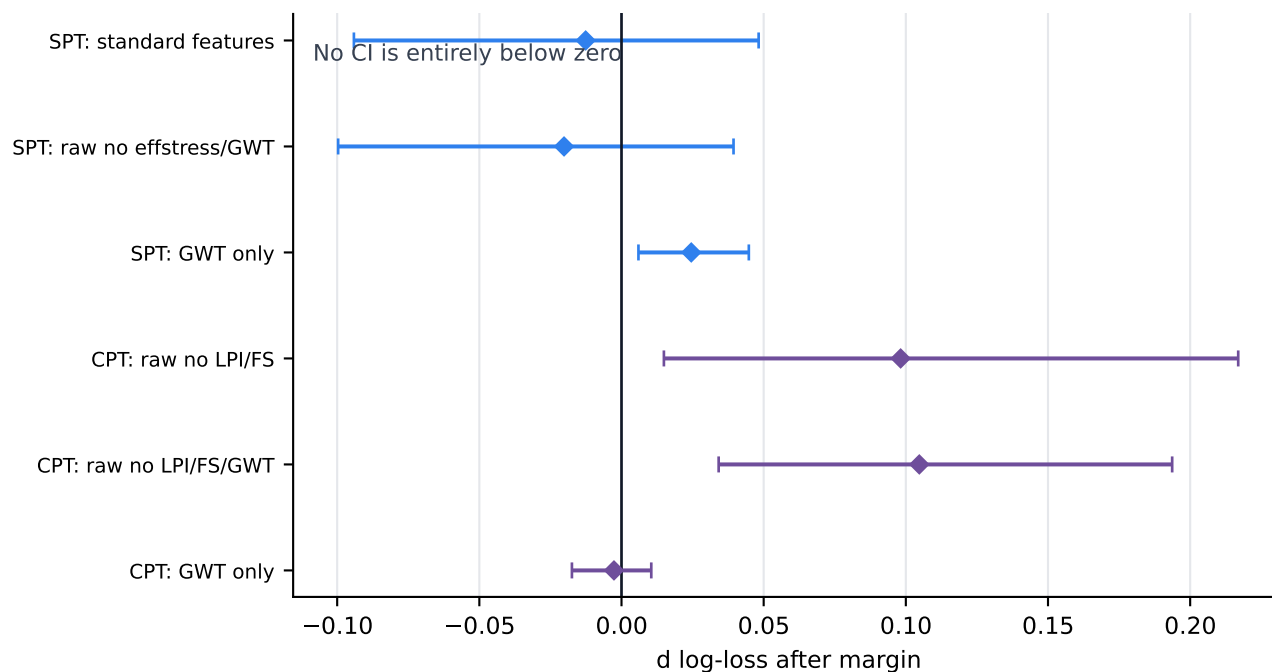
A Random validation inflates apparent ML skill



B Full models do not improve OOS likelihood



C Residual/raw variables after the margin



D Leave-one-region-out transfer

